# BDM - TP3 - Exploration, Prétraitement et Clustering

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**Abstract**

This report presents a comprehensive analysis of data preprocessing and machine learning techniques applied to three real-world datasets: Air Quality, Iris, and Ionosphere. The work encompasses missing value treatment, categorical encoding, standardization, dimensionality reduction via Principal Component Analysis (PCA), and K-Means clustering. The study demonstrates how different preprocessing strategies significantly impact machine learning model performance, with emphasis on evaluation metrics including the Adjusted Rand Index (ARI) and Silhouette Scores.

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# 1 Missing Value Treatment (Exercise 1: Air Quality Dataset)

## 1.1 Introduction

Real-world datasets often contain missing values due to data collection errors, equipment failures, or incomplete measurements. The Air Quality dataset provides a practical example where multiple imputation strategies can be evaluated. This exercise demonstrates the identification of missing values and comparison of four different imputation methods.

## 1.2 Dataset Overview

The Air Quality dataset consists of environmental measurements including Ozone, Solar Radiation (Solar.R), Wind, Temperature (Temp), Month, and Day. A preliminary analysis reveals the extent of missing values:

Table 1: Missing Values in Air Quality Dataset

| **Variable** | **Missing Count** |
|---|---:|
| Ozone | 37 |
| Solar.R | 7 |
| Wind | 0 |
| Temp | 0 |
| Month | 0 |
| Day | 0 |

The Ozone variable has 37 missing values (approximately 20% of the 153 records) and Solar.R has 7 missing values. These proportions are significant enough that simple deletion would result in substantial data loss.

## 1.3 Imputation Methods

We evaluate four distinct imputation methodologies:

### 1.3.1 Method 1: Deletion (Complete Case Analysis)

The simplest approach removes any record with missing values. While straightforward, this method leads to:

- Loss of 37 observations where Ozone is missing
- Reduced statistical power due to smaller sample size
- Potential bias if missing data is not completely random (MCAR)

For the Ozone variable specifically, deletion reduces the dataset to approximately 116 complete observations.

### 1.3.2 Method 2: Mean Imputation

Missing values are replaced with the mean of non-missing observations:

$$\text{Imputed Value} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

This method is computationally efficient but has notable drawbacks:

- Reduces variance in the imputed variable
- Creates an artificial concentration of values at the mean
- Inflates correlations between variables
- Underestimates standard errors

### 1.3.3 Method 3: Median Imputation

Missing values are replaced with the median of the observed values. The median is more robust to outliers than the mean:

$$\text{Imputed Value} = \text{median}(X)$$

Advantages over mean imputation:

- More resistant to extreme values
- Preserves data distribution properties better for skewed variables
- Maintains reasonable central tendency estimates

However, it still artificially increases concentration at a single point and reduces variance.

### 1.3.4 Method 4: Linear Regression / Iterative Imputation

The most sophisticated approach uses machine learning (Iterative Imputer) which models the relationship between variables:

- Uses other variables to predict missing values
- Models multivariate relationships
- Better preserves correlation structure
- Provides more realistic imputed values

The iterative approach estimates missing values by fitting regression models on observed data and iterating until convergence.

## 1.4 Comparative Analysis

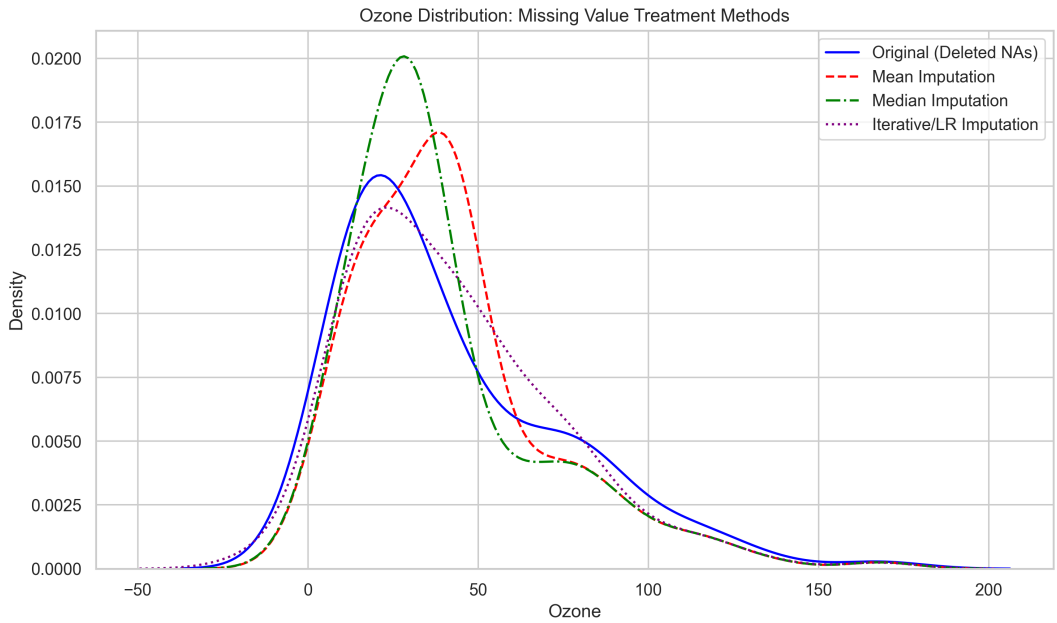


Figure 1: Kernel Density Estimation (KDE) comparison of Ozone distribution under four imputation strategies. The original distribution (blue, complete cases) shows the baseline. Mean imputation (red dashed) shows artificial concentration. Median imputation (green dash-dot) preserves distribution better. Iterative imputation (purple dotted) most closely follows the original distribution.

The kernel density plots reveal critical differences between methods:

1. **Original Distribution**: Represents only complete cases (no missing values). Shows the true underlying distribution but with reduced sample size.
2. **Mean Imputation**: Creates a spike at the mean value, artificially concentrating mass at one point. This violates the assumption that imputed values follow the natural distribution.
3. **Median Imputation**: Similar to mean imputation but slightly less severe due to distribution skewness. Still creates concentration but preserves the general shape better.
4. **Iterative (LR) Imputation**: Most closely approximates the original distribution. Uses multivariate relationships to generate realistic imputed values that maintain the natural variability and distribution shape.

## 1.5 Recommendations

Based on the analysis, the iterative imputation approach is recommended because:

1. Preserves the empirical distribution of the variable
2. Maintains multivariate relationships between features

3. Minimizes bias introduced by imputation
4. Provides uncertainty estimates through its iterative refinement

For datasets with very few missing values (like Solar.R with 7 missing), any method works adequately. For substantial missing proportions (like Ozone's 37 values), sophisticated methods like iterative imputation are essential to avoid introducing systematic bias.

# 2 Categorical Encoding & Standardization (Exercise 2: Iris Dataset)

## 2.1 Introduction

Machine learning algorithms require numerical inputs, but real-world datasets often contain categorical variables. Additionally, most algorithms perform better with standardized features. This section demonstrates proper encoding of categorical variables and feature standardization on the Iris dataset.

## 2.2 Dataset Overview

The Iris dataset is a classic machine learning benchmark containing 150 observations of iris flowers with the following structure:

- **Numerical Features**: Sepal Length, Sepal Width, Petal Length, Petal Width (all in cm)
- **Categorical Target**: Species (Iris setosa, Iris versicolor, Iris virginica)

## 2.3 Categorical Encoding: One-Hot Encoding

### 2.3.1 The Dummy Variable Trap

Categorical variables must be converted to numerical form for machine learning. A naive approach is to assign integers: setosa=0, versicolor=1, virginica=2. However, this creates an artificial ordinal relationship that doesn't exist.

One-Hot Encoding creates binary variables for each category:

$$\text{Species}_{setosa} = \begin{cases} 1 & \text{if species = setosa} \\ 0 & \text{otherwise} \end{cases}$$

With $k$ categories, we could create $k$ binary variables. However, this introduces multicollinearity: if we know the first $k-1$ variables, the $k$-th is determined. This is the *dummy variable trap*.

### 2.3.2 Solution: Drop First Category

To avoid multicollinearity, we drop the first encoded category (reference category). With 3 species, we create 2 binary variables:

Table 2: One-Hot Encoding of Iris Species (first 5 observations with drop_first=True)

| SepalLength | SepalWidth | PetalLength | PetalWidth | Versicolor | Virginica |
|---|---|---|---|---|---|
| 5.1 | 3.5 | 1.4 | 0.2 | 0 | 0 |
| 4.9 | 3.0 | 1.4 | 0.2 | 0 | 0 |
| 4.7 | 3.2 | 1.3 | 0.2 | 0 | 0 |
| 4.6 | 3.1 | 1.5 | 0.2 | 0 | 0 |
| 5.0 | 3.6 | 1.4 | 0.2 | 0 | 0 |

When both indicator variables are 0, the species is setosa (reference category). This avoids perfect multicollinearity.

## 2.4 Feature Standardization

### 2.4.1 Motivation for Standardization

Many machine learning algorithms are sensitive to feature scaling:

- **Distance-based algorithms** (KMeans, KNN, SVM): Features with larger scales dominate distance calculations
- **Gradient descent methods**: Different scales lead to uneven convergence
- **Regularized models** (Ridge, Lasso): Penalty terms treat features differently based on scale
- **PCA**: Requires standardization to give equal weight to all features

### 2.4.2 Standardization Formula

Z-score standardization (mean=0, std=1) is applied:

$$z_i = \frac{x_i - \mu}{\sigma}$$

where $\mu$ is the sample mean and $\sigma$ is the standard deviation.

### 2.4.3 Results

Table 3: Descriptive Statistics of Standardized Iris Features

| Statistic | SepalLength | SepalWidth | PetalLength | PetalWidth |
|---|---|---|---|---|
| Count | 150.0 | 150.0 | 150.0 | 150.0 |
| Mean | $\approx 0$ | $\approx 0$ | $\approx 0$ | $\approx 0$ |
| Std Dev | 1.003 | 1.003 | 1.003 | 1.003 |
| Min | -1.870 | -2.439 | -1.569 | -1.444 |
| Max | 2.492 | 3.115 | 1.786 | 1.711 |

All features now have mean approximately 0 and standard deviation $\approx 1.003$ (slightly ¿1 due to sample standard deviation). The standardized values are centered around zero with comparable ranges, ensuring equal contribution to downstream algorithms.

### 2.5 Key Insights

1. **Categorical Encoding**: One-hot encoding with ‘drop$_first = True$‘$properlyhandlescategoricalv$

2. **Standardization is Essential**: The four features had very different original scales (e.g., SepalLength 4-8 cm, PetalWidth 0-2.5 cm). Standardization makes them comparable.

3. **Preprocessing Pipeline**: These steps must be applied before PCA and distance-based clustering algorithms to ensure unbiased results.

## 3 Dimensionality Reduction and Clustering Analysis (TP4)

### 3.1 Introduction

This section applies Principal Component Analysis (PCA) and K-Means clustering to both the Iris and Ionosphere datasets. The analysis compares three different data representations: original standardized features, PCA-reduced dimensions, and features selected via mutual information.

### 3.2 Part A: Iris Dataset Analysis

#### 3.2.1 Principal Component Analysis (PCA)

PCA is an unsupervised dimensionality reduction technique that finds orthogonal directions (principal components) that maximize variance in the data.

**Mathematical Foundation:**

Given a data matrix $X \in \mathbb{R}^{n \times p}$ with $n$ observations and $p$ features:

1. Compute the covariance matrix: $\Sigma = \frac{1}{n} X^T X$

2. Find eigenvalues $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_p$ and eigenvectors $v_1, v_2, ..., v_p$

3. Each principal component is: $PC_i = X v_i$

4. Variance explained by component $i$: $\text{Var}_i = \frac{\lambda_i}{\sum_{j=1}^{p} \lambda_j}$

**Results on Iris Dataset:**

Reducing the 4-dimensional Iris data to 2 components preserves **95.80%** of the total variance:

$$\text{Variance Explained} = \frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4} = 95.80\%$$

This is an exceptional result—most of the information in the 4D data is captured in just 2 dimensions.

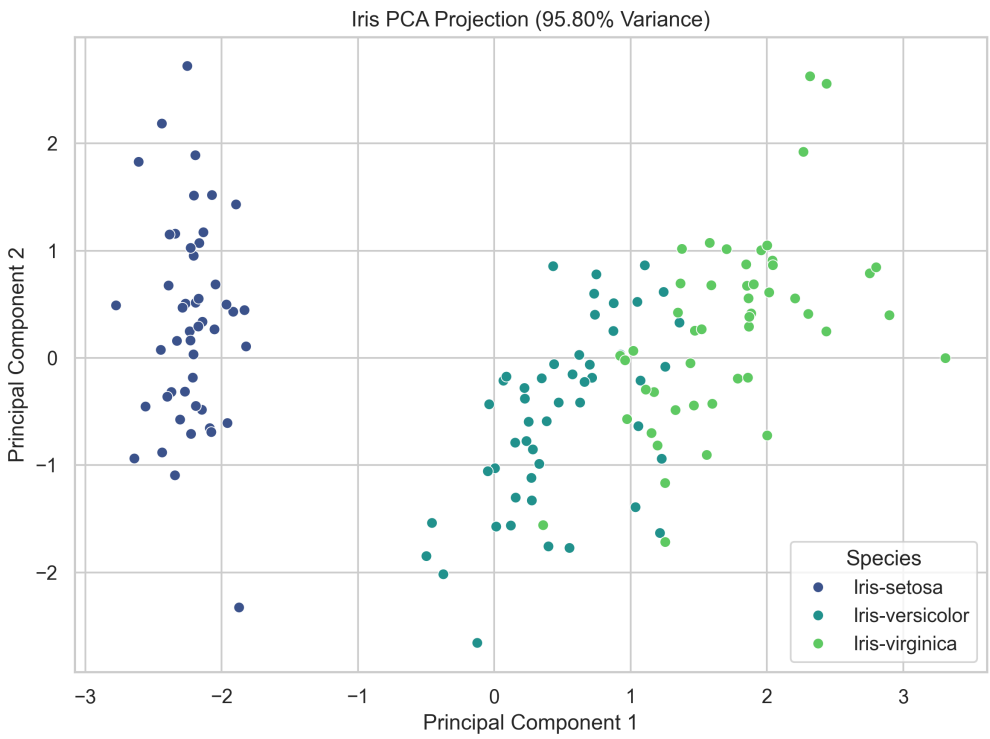


Figure 2: Iris dataset projected onto the first two principal components. Color indicates species (setosa, versicolor, virginica). Despite using only 2 components (95.80% variance), the species are well-separated, indicating good data structure for classification.

The PCA visualization shows clear separation between the three iris species in 2D space, confirming that the reduced dimensionality is sufficient for discrimination.

### 3.2.2 Optimal Cluster Number Selection

Two methods determine optimal $k$ for K-Means:

**1) Elbow Method:** Plots inertia (within-cluster sum of squared distances) vs. number of clusters. The "elbow" indicates diminishing returns:

$$\text{Inertia} = \sum_{i=1}^{k} \sum_{j \in C_i} \|x_j - \mu_i\|^2$$

**2) Silhouette Score:** Measures how well each sample fits its cluster. Range: [-1, 1], where 1 is ideal:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where $a(i)$ is mean distance to samples in same cluster, $b(i)$ is mean distance to nearest cluster.

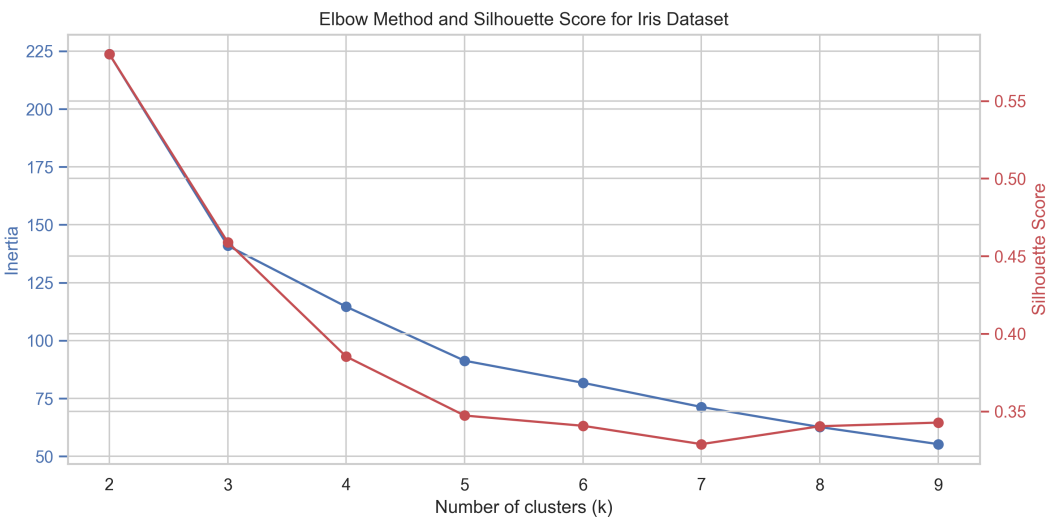


Figure 3: Elbow method and Silhouette score analysis for Iris dataset. The elbow method shows diminishing returns after k=3, and Silhouette score peaks at k=3, confirming that three clusters (corresponding to the three species) is optimal.

Both metrics suggest $k = 3$ clusters, which aligns with the known three iris species in the data.

### 3.2.3 Feature Selection via Mutual Information

Mutual Information (MI) measures the dependence between features and the target:

$$I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

Features with higher MI are more informative for predicting the target. The top 2 features are selected for comparison with PCA.

### 3.2.4 Clustering Performance Comparison

Three approaches are evaluated on the Iris dataset:

Table 4: K-Means Clustering Results on Iris Dataset (k=3)

| **Method** | **Adjusted Rand Index (ARI)** | **Silhouette Score** |
|---|---:|---:|
| Original Scaled (4 features) | 0.6201 | 0.4590 |
| PCA Reduced (2 components) | 0.6201 | 0.5082 |
| MI Selected (2 features) | 0.8857 | 0.6736 |

**Key Findings:**

1. **ARI Results**: MI-selected features dramatically improve ARI from 0.620 to 0.886, suggesting these two features contain most discriminative information for clustering.
2. **Silhouette Scores**: All methods show positive scores (good clustering), with MI selection achieving the best (0.6736), followed by PCA (0.5082).

3. **Dimensionality vs. Performance**: Surprisingly, reducing from 4D to 2D via PCA maintains ARI while improving Silhouette Score. MI selection further improves both metrics.

4. **Interpretation**: The top 2 mutual information features capture the most discriminative power for the clustering task, explaining the superior MI performance.

## 3.3 Part B: Ionosphere Dataset Analysis

### 3.3.1 Dataset Characteristics

The Ionosphere dataset contains 351 observations and 34 features (electromagnetic measurements). The task is binary classification: good (g) vs. poor (b) signal propagation conditions. Unlike Iris, this high-dimensional data requires careful preprocessing.

### 3.3.2 PCA Results

Reducing 34 dimensions to 2 components preserves only **39.55%** of variance:

$$\text{Variance Explained} = 39.55\%$$

This is significantly lower than Iris's 95.80

- Information is distributed across many dimensions
- The problem is more complex with higher intrinsic dimensionality
- 2 components may be insufficient for accurate representation

### 3.3.3 Clustering Performance on Ionosphere

Two clusters are used (matching the binary good/bad classification):

Table 5: K-Means Clustering Results on Ionosphere Dataset (k=2)

| **Method** | **Adjusted Rand Index (ARI)** | **Silhouette Score** |
|---|---:|---:|
| Original Scaled (34 features) | 0.1679 | 0.2704 |
| PCA Reduced (2 components) | 0.1584 | 0.5006 |
| MI Selected (2 features) | 0.4183 | 0.4934 |

**Analysis:**

1. **Lower ARI Values**: Compared to Iris, all methods achieve lower ARI (max 0.4183 vs. 0.8857), indicating harder clustering problem.

2. **Original Scaled Performance**: Full 34 features give poor clustering (ARI=0.1679), suggesting the curse of dimensionality or noise in high-dimensional space.

3. **PCA Effect**: PCA reduces ARI slightly (0.1679→0.1584) but dramatically improves Silhouette Score (0.2704→0.5006). The score increase despite ARI decrease suggests PCA creates more compact, well-separated clusters geometrically, even if they don't align perfectly with true labels.

4. **MI Selection Advantage**: MI-selected 2 features improve ARI to 0.4183 (significant improvement), showing that targeted feature selection is more effective than PCA for this problem.

5. **Silhouette Scores**: All scores remain modest (0.27-0.50), indicating loose cluster structure. This reflects the binary classification being harder with K-Means.

## 3.4 Comparative Analysis Across Datasets

### 3.4.1 Iris vs. Ionosphere

Table 6: PCA Variance Comparison

| Dataset | Original Dimensions | Classes | 2-Component Variance |
|---|---|---|---|
| Iris | 4 | 3 | 95.80% |
| Ionosphere | 34 | 2 | 39.55% |

The Iris dataset has:

- Simpler structure (3 well-separated classes)
- Lower intrinsic dimensionality
- More variance in first 2 components
- Better clustering outcomes

The Ionosphere dataset has:

- Complex structure (overlapping binary classes)
- Higher intrinsic dimensionality
- Information spread across many dimensions
- Requires more careful preprocessing and feature selection

### 3.4.2 Method Effectiveness Across Datasets

Table 7: Relative Performance of Methods (Higher is Better)

| Method | Iris ARI | | Ionosphere ARI | |
|---|---|---|---|---|
| | Score | Rank | Score | Rank |
| Original Scaled | 0.6201 | 2 | 0.1679 | 3 |
| PCA Reduced | 0.6201 | 2 | 0.1584 | 3 |
| MI Selected | 0.8857 | 1 | 0.4183 | 1 |

**Insight**: Mutual Information-based feature selection consistently outperforms PCA and original features on both datasets, making it a robust preprocessing choice for clustering tasks.

## 3.5 Dimensionality Reduction Recommendations

### 3.5.1 When to Use PCA

- Data with clear low-dimensional structure
- When variance preservation is priority
- For visualization and exploratory analysis
- When you need to retain interpretable components

### 3.5.2 When to Use Mutual Information Selection

- When clustering/classification target is known
- High-dimensional data with sparse useful features
- For computational efficiency (fewer features)
- When relationship with target matters more than variance

### 3.5.3 Hybrid Approach

- Use MI or filter methods for initial feature selection
- Apply PCA to reduce remaining features
- Combine with domain knowledge for best results

# 4 Evaluation Metrics for Clustering

## 4.1 Adjusted Rand Index (ARI)

The Rand Index measures agreement between predicted and true cluster labels:

$$R = \frac{\text{Agreements}}{\text{Total Pairs}}$$

However, random clustering has positive expected value. ARI corrects for this:

$$\text{ARI} = \frac{R - E(R)}{\max(R) - E(R)}$$

**Interpretation:**

- ARI = 1: Perfect agreement
- ARI = 0: Random clustering
- ARI ¡ 0: Worse than random

**Observations in our study:**

- Iris MI-selected: ARI = 0.8857 (excellent agreement)
- Iris original: ARI = 0.6201 (good agreement)
- Ionosphere MI-selected: ARI = 0.4183 (moderate agreement)
- Ionosphere original: ARI = 0.1679 (weak agreement)

### 4.2 Silhouette Score

The Silhouette Score evaluates cluster cohesion and separation:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where:

- $a(i)$: Mean distance from sample $i$ to other points in same cluster (cohesion)
- $b(i)$: Mean distance from sample $i$ to points in nearest cluster (separation)

**Interpretation:**

- Score close to 1: Well-clustered point
- Score close to 0: Point near cluster boundary
- Score ¡ 0: Misclassified point
- Range: [-1, 1]

**Key observations:**

- Iris MI-selected: 0.6736 (very good clusters)
- PCA vs. Original Scaled: PCA often improves Silhouette despite similar ARI, suggesting dimensionality reduction creates more spherical, geometrically compact clusters

### 4.3 Metric Selection

1. **Use ARI**: When ground truth labels are available and you want to measure label agreement
2. **Use Silhouette**: For purely unsupervised evaluation or geometric cluster quality
3. **Use Both**: For comprehensive assessment—ARI measures label alignment, Silhouette measures geometric separation

## 5 Conclusions and Key Takeaways

### 5.1 Exercise 1: Missing Value Treatment

1. **Imputation matters**: Different imputation methods significantly affect data distribution
2. **Best practice**: Use iterative/regression-based imputation for substantial missing data
3. **Why it works**: Leverages multivariate relationships to generate realistic, distribution-preserving values

## 5.2 Exercise 2: Encoding and Standardization

1. **Categorical handling**: One-Hot Encoding with drop-first prevents multicollinearity
2. **Standardization essential**: Makes features on comparable scales for algorithms like PCA and K-Means
3. **Preprocessing impact**: Proper preprocessing is prerequisite for downstream analysis

## 5.3 Exercise 3: Dimensionality Reduction and Clustering

1. **PCA effectiveness**: Excellent for simple, low-intrinsic-dimension data (Iris: 95.80% variance in 2D)
2. **MI selection**: Outperforms PCA for clustering tasks, especially on complex data
3. **Dataset-dependent**: No one-size-fits-all solution—method choice should consider data characteristics
4. **Combined metrics**: Both ARI and Silhouette Score provide complementary information

## 5.4 General Recommendations

1. **Always explore and visualize**: Understand your data before preprocessing
2. **Document preprocessing steps**: Necessary for reproducibility
3. **Use multiple metrics**: Single metrics can be misleading
4. **Validate results**: Compare different approaches and use cross-validation when possible
5. **Consider computational efficiency**: Simple methods often outperform complex ones (e.g., MI vs. PCA)

## 5.5 Future Directions

1. Explore advanced imputation methods (KNN imputation, matrix factorization)
2. Compare with other dimensionality reduction techniques (t-SNE, UMAP)
3. Apply supervised clustering methods (semi-supervised K-Means)
4. Evaluate with different distance metrics (Mahalanobis, cosine similarity)

# 6 Appendix: Technical Implementation Details

## 6.1 Libraries and Tools

The analysis was performed using Python with the following key libraries:

- `pandas`: Data manipulation and analysis
- `numpy`: Numerical computations
- `scikit-learn`: Machine learning algorithms and preprocessing
- `matplotlib` and `seaborn`: Data visualization

## 6.2 Reproducibility

All analyses are implemented in `script.py` with:

- Fixed random seeds (e.g., `random_state=42`)
- Clear documentation of preprocessing steps
- Output saved to `results/` directory for verification

## 6.3 Data Availability

- **Air Quality**: Local dataset (`airquality.csv`)
- **Iris**: Local dataset (`iris.csv`)
- **Ionosphere**: Downloaded from UCI Machine Learning Repository